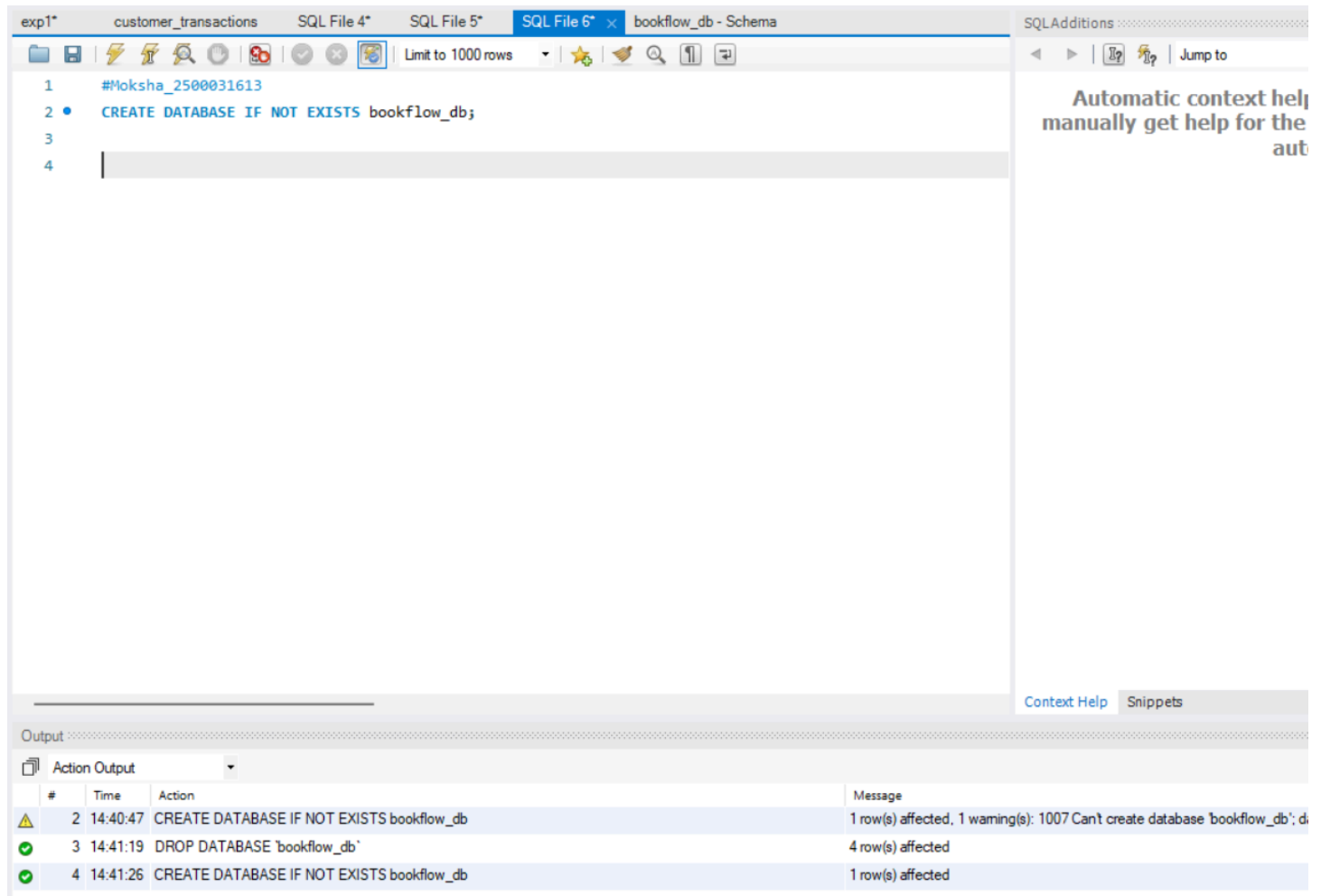


2500031613_Moksha Ratna

WEEK-1 PRACTICAL DATABASE SYSTEM ENGINEERING

Query : 1 CREATE DATABASE IF NOT EXISTS bookflow_db;

output:



The screenshot shows a SQL IDE interface with a query editor and an output window. The query editor contains the following SQL statement:

```
1 #Moksha_2500031613
2 CREATE DATABASE IF NOT EXISTS bookflow_db;
3
4
```

The output window displays the results of the query execution:

#	Time	Action	Message
2	14:40:47	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bookflow_db'; database doesn't exist
3	14:41:19	DROP DATABASE 'bookflow_db'	4 row(s) affected
4	14:41:26	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected

Query 2: USE bookflow_db;

output:

The screenshot shows a SQL IDE with a script containing three queries. The output pane displays the execution results for these queries.

SQL Script:

```
1 #Moksha_2500031613
2 CREATE DATABASE IF NOT EXISTS bookflow_db;
3 USE bookflow_db;
```

Output:

#	Time	Action	Message
3	14:41:19	DROP DATABASE 'bookflow_db'	4 row(s) affected
4	14:41:26	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected
5	14:48:14	USE bookflow_db	0 row(s) affected

Query 3: CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL, isbn VARCHAR(13) NOT NULL UNIQUE, published_year INT, CONSTRAINT chk_published_year CHECK (published_year < 2027));

output:

The screenshot shows a SQL IDE with a script editor and an output pane. The script editor contains the following SQL code:

```

1  #Moksha_2500031613
2  • CREATE DATABASE IF NOT EXISTS bookflow_db;
3  • USE bookflow_db;
4  • CREATE TABLE books (
5      book_id
6      INT AUTO_INCREMENT PRIMARY KEY,
7      title
8      VARCHAR(255) NOT NULL,
9      isbn
10     VARCHAR(13) NOT NULL UNIQUE,
11     published_year INT,
12     CONSTRAINT chk_published_year CHECK (published_year < 2027)
13 );
14
15

```

The output pane shows the execution results of the first three commands:

#	Time	Action	Message
4	14:41:26	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected
5	14:48:14	USE bookflow_db	0 row(s) affected
6	14:49:23	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) ...	0 row(s) affected

Query 4: Create the members table CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(100) NOT NULL, email VARCHAR(150) NOT NULL UNIQUE);

output:

exp1* customer_transactions SQL File 4* SQL File 5* SQL File 6* x bookflow_db - Schema SQLAdditions

Limit to 1000 rows

```
1 #Moksha_2500031613
2 • CREATE DATABASE IF NOT EXISTS bookflow_db;
3 • USE bookflow_db;
4 • CREATE TABLE books (
5     book_id
6     INT AUTO_INCREMENT PRIMARY KEY,
7     title
8     VARCHAR(255) NOT NULL,
9     isbn
10    VARCHAR(13) NOT NULL UNIQUE,
11    published_year INT,
12    CONSTRAINT chk_published_year CHECK (published_year < 2027)
13 );
14 • CREATE TABLE members (
15     member_id INT AUTO_INCREMENT PRIMARY KEY,
16     full_name VARCHAR(100) NOT NULL,
17     email
18     VARCHAR(150) NOT NULL UNIQUE
19 );
20
```

Context Help

Output

Action Output

#	Time	Action	Message
✓ 5	14:48:14	USE bookflow_db	0 row(s) affected
✓ 6	14:49:23	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) ...	0 row(s) affected
✓ 7	14:50:43	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(100)...	0 row(s) affected

Query 5 (verification): Inspect the table structures DESCRIBE members;
output:

```

16 full_name VARCHAR(100) NOT NULL,
17 email
18 VARCHAR(150) NOT NULL UNIQUE
19 );
20 • DESCRIBE members;

```

Field	Type	Null	Key	Default	Extra
member_id	int	NO	PRI	NULL	auto_increment
full_name	varchar(100)	NO		NULL	
email	varchar(150)	NO	UNI	NULL	

Result 1 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message
6	14:49:23	CREATE TABLE books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255)...	0 row(s) affected
7	14:50:43	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(10...	0 row(s) affected
8	14:51:40	DESCRIBE members	3 row(s) returned

Query 6: Insert 3 books INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean Code', '9780132350884', 2008), ('Atomic Habits', '9780735211292', 2018);

output:

```
3 • USE bookflow_db;
4 • CREATE TABLE books (
5     book_id
6     INT AUTO_INCREMENT PRIMARY KEY,
7     title
8     VARCHAR(255) NOT NULL,
9     isbn
10    VARCHAR(13) NOT NULL UNIQUE,
11    published_year INT,
12    CONSTRAINT chk_published_year CHECK (published_year < 2027)
13 );
14 • CREATE TABLE members (
15     member_id INT AUTO_INCREMENT PRIMARY KEY,
16     full_name VARCHAR(100) NOT NULL,
17     email
18     VARCHAR(150) NOT NULL UNIQUE
19 );
20 • DESCRIBE members;
21 • INSERT INTO books (title, isbn, published_year) VALUES
22 ('The Alchemist', '9780061122415', 1988),
23 ('Clean Code', '9780132350884', 2008),
24 ('Atomic Habits', '9780735211292', 2018);
25
```

Output:

#	Time	Action	Message
7	14:50:43	CREATE TABLE members (member_id INT AUTO_INCREMENT PRIMARY KEY, full_name VARCHAR(10...	0 row(s) affected
8	14:51:40	DESCRIBE members	3 row(s) returned
9	14:52:43	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Query 7: Insert 3 members INSERT INTO members (full_name, email)
VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Sharma',
'priya.sharma@example.com'), ('Ravi Verma', 'ravi.verma@example.com');
output:

Limit to 1000 rows

```

6  INT AUTO_INCREMENT PRIMARY KEY,
7  title
8  VARCHAR(255) NOT NULL,
9  isbn
10 VARCHAR(13) NOT NULL UNIQUE,
11 published_year INT,
12 CONSTRAINT chk_published_year CHECK (published_year < 2027)
13 );
14 CREATE TABLE members (
15 member_id INT AUTO_INCREMENT PRIMARY KEY,
16 full_name VARCHAR(100) NOT NULL,
17 email
18 VARCHAR(150) NOT NULL UNIQUE
19 );
20 DESCRIBE members;
21 INSERT INTO books (title, isbn, published_year) VALUES
22 ('The Alchemist', '9780061122415', 1988),
23 ('Clean Code', '9780132350884', 2008),
24 ('Atomic Habits', '9780735211292', 2018);
25 INSERT INTO members (full_name, email) VALUES
26 ('Anil Kumar', 'anil.kumar@example.com'),
27 ('Priya Sharma', 'priya.sharma@example.com'),
28 ('Ravi Verma', 'ravi.verma@example.com');

```

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Context Help Snippets

Output

Action Output

#	Time	Action	Message
8	14:51:40	DESCRIBE members	3 row(s) returned
9	14:52:43	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
10	14:53:38	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Sharma...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0

Test 1: Duplicate ISBN (UNIQUE violation) INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000);

output:

```

20 DESCRIBE members;
21 INSERT INTO books (title, isbn, published_year) VALUES
22 ('The Alchemist', '9780061122415', 1988),
23 ('Clean Code', '9780132350884', 2008),
24 ('Atomic Habits', '9780735211292', 2018);
25 INSERT INTO members (full_name, email) VALUES
26 ('Anil Kumar', 'anil.kumar@example.com'),
27 ('Priya Sharma', 'priya.sharma@example.com'),
28 ('Ravi Verma', 'ravi.verma@example.com');
29
30 INSERT INTO books (title, isbn, published_year)
31 VALUES ('Fake Copy', '9780061122415', 2000);

```

Context Help Snippets

Output

Action Output

#	Time	Action	Message
9	14:52:43	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '9780061122415', 1988), ('Clean ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
10	14:53:38	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Sharma...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
11	14:54:43	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'

Test 2: NULL title (NOT NULL violation) INSERT INTO books (title, isbn, published_year) VALUES (NULL, '99999999999999', 2010);

output:

exp1* customer_transactions SQL File 4* SQL File 5* SQL File 6* bookflow_db - Schema SQLAdditions

Limit to 1000 rows

```

12  CONSTRAINT chk_published_year CHECK (published_year < 2027)
13  );
14  CREATE TABLE members (
15  member_id INT AUTO_INCREMENT PRIMARY KEY,
16  full_name VARCHAR(100) NOT NULL,
17  email
18  VARCHAR(150) NOT NULL UNIQUE
19  );
20  DESCRIBE members;
21  INSERT INTO books (title, isbn, published_year) VALUES
22  ('The Alchemist', '9780061122415', 1988),
23  ('Clean Code', '9780132350884', 2008),
24  ('Atomic Habits', '9780735211292', 2018);
25  INSERT INTO members (full_name, email) VALUES
26  ('Anil Kumar', 'anil.kumar@example.com'),
27  ('Priya Sharma', 'priya.sharma@example.com'),
28  ('Ravi Verma', 'ravi.verma@example.com');
29
30  INSERT INTO books (title, isbn, published_year)
31  VALUES ('Fake Copy', '9780061122415', 2000);
32
33  INSERT INTO books (title, isbn, published_year)
34  VALUES (NULL, '9999999999999', 2010);

```

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Context Help Snippets

Output

Action Output

#	Time	Action	Message
✓ 10	14:53:38	INSERT INTO members (full_name, email) VALUES ('Anil Kumar', 'anil.kumar@example.com'), ('Priya Sharma...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0
✗ 11	14:54:43	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'boo
✗ 12	14:55:52	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010)	Error Code: 1048. Column 'title' cannot be null

Test 3: Future publication year (CHECK violation) INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '88888888888888', 2030);

output:

The screenshot shows a SQL IDE with a script containing several SQL statements. The script includes a table creation for 'members', a description, and three insert statements for the 'books' table. The third insert statement is highlighted in blue. The output window at the bottom shows three error messages corresponding to the insert statements.

```
14 CREATE TABLE members (  
15     member_id INT AUTO_INCREMENT PRIMARY KEY,  
16     full_name VARCHAR(100) NOT NULL,  
17     email  
18     VARCHAR(150) NOT NULL UNIQUE  
19 );  
20 DESCRIBE members;  
21 INSERT INTO books (title, isbn, published_year) VALUES  
22 ('The Alchemist', '9780061122415', 1988),  
23 ('Clean Code', '9780132350884', 2008),  
24 ('Atomic Habits', '9780735211292', 2018);  
25 INSERT INTO members (full_name, email) VALUES  
26 ('Anil Kumar', 'anil.kumar@example.com'),  
27 ('Priya Sharma', 'priya.sharma@example.com'),  
28 ('Ravi Verma', 'ravi.verma@example.com');  
29  
30 INSERT INTO books (title, isbn, published_year)  
31 VALUES ('Fake Copy', '9780061122415', 2000);  
32  
33 INSERT INTO books (title, isbn, published_year)  
34 VALUES (NULL, '9999999999999', 2010);  
35 INSERT INTO books (title, isbn, published_year)  
36 VALUES ('Time Traveler', '8888888888888', 2030);
```

Output:

#	Time	Action	Message
11	14:54:43	INSERT INTO books (title, isbn, published_year) VALUES ('Fake Copy', '9780061122415', 2000)	Error Code: 1062. Duplicate entry '9780061122415' for key 'books.isbn'
12	14:55:52	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010)	Error Code: 1048. Column 'title' cannot be null
13	14:56:34	INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '8888888888888', 2030)	Error Code: 3819. Check constraint 'chk_published_year' is violated.

Test 4: Duplicate email (UNIQUE violation) INSERT INTO members (full_name, email) VALUES ('Anil Clone', 'anil.kumar@example.com');

output:

exp1*customer_transactionsSQL File 4*SQL File 5*SQL File 6*bookflow_db - Schema

Limit to 1000 rows

SQLAdditions

16full_name VARCHAR(100) NOT NULL,
17email
18VARCHAR(150) NOT NULL UNIQUE
19);
20• DESCRIBE members;
21• INSERT INTO books (title, isbn, published_year) VALUES
22('The Alchemist', '9780061122415', 1988),
23('Clean Code', '9780132350884', 2008),
24('Atomic Habits', '9780735211292', 2018);
25• INSERT INTO members (full_name, email) VALUES
26('Anil Kumar', 'anil.kumar@example.com'),
27('Priya Sharma', 'priya.sharma@example.com'),
28('Ravi Verma', 'ravi.verma@example.com');
29
30• INSERT INTO books (title, isbn, published_year)
31VALUES ('Fake Copy', '9780061122415', 2000);
32
33• INSERT INTO books (title, isbn, published_year)
34VALUES (NULL, '9999999999999', 2010);
35• INSERT INTO books (title, isbn, published_year)
36VALUES ('Time Traveler', '8888888888888', 2030);
37• INSERT INTO members (full_name, email)
38VALUES ('Anil Clone', 'anil.kumar@example.com');

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Context HelpSnippets

Output

Action Output

#	Time	Action	Message
✖12	14:55:52	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010)	Error Code: 1048. Column 'title' cannot be null
✖13	14:56:34	INSERT INTO books (title, isbn, published_year) VALUES ('Time Traveler', '8888888888888', 2030)	Error Code: 3819. Check constraint 'chk_published_year' is violated.
✖14	14:57:06	INSERT INTO members (full_name, email) VALUES ('Anil Clone', 'anil.kumar@example.com')	Error Code: 1062. Duplicate entry 'anil.kumar@example.com' for key 'members.email'